Vathaluru Akhileswar Reddy

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Profile Summary

- Backend engineer specializing in building production-grade systems with FastAPI and Spring Boot, featuring domain-driven design, microservices architecture, and serverless AWS deployments.
- Expert in AI agent development using LangGraph and LangChain for conversational AI with multi-turn dialogue management, custom tool integration, and vector databases.
- Proven track record of designing scalable architectures including event-driven systems with RabbitMQ, RESTful APIs with OAuth2/JWT authentication, and CI/CD pipelines achieving 80%+ test coverage.

Technical Skills

Languages: Python, Go, Java, JavaScript, C, C++
Backend Frameworks: FastAPI, Spring Boot, Express.js
AI/LLM: LangGraph, LangChain, OpenAI API, Tavily, Inngest Agent Kit
Frontend: React, Tailwind CSS, Vite
Microservices: Spring Cloud (Gateway, Eureka, Config), RabbitMQ
Databases: PostgreSQL, MongoDB, MySQL, Redis, Pinecone (Vector DB)
Cloud & DevOps: AWS (Lambda, S3, SES, Cognito), AWS CDK, Docker, GitHub Actions
ORM: SQLAlchemy, Spring Data JPA, Mongoose
Auth & Payments: OAuth2, JWT, AWS Cognito, Clerk, Razorpay
Testing: Pytest, Ruff, Mypy, JUnit

Education

KL University — **Guntur, India**
Bachelor of Technology in Computer Science and Engineering, CGPA: 9.43 — *2021–2025*

Experience

Ideyalabs — **Hyderabad, India**
Backend Developer — *2025–Present*

APEX – Transportation Back Office System

- Developed a production-grade transportation back-office backend using FastAPI, PostgreSQL, SQLAlchemy, Pydantic, and domain-driven module architecture.
- Built client workflows for account lookup, shipment creation/listing/export, pickup scheduling and rescheduling, quick quotes, rater calculation, freight density calculation, ZIP search, instruction codes, and lookup APIs.
- Implemented Pricing & Rating modules for fuel prices, fuel scales, accessorial charges/schedules, rate bases, lane minimums, high-cost ZIPs, global rates, pricing rules, and volume rates.
- Integrated SMC3 freight rating, HERE geocoding, EIA fuel pricing, Daylight services, Falvey insurance, AWS S3, AWS SES, and AWS Secrets Manager.
- Built document workflows for bill of lading PDFs, shipping labels, blind shipment forms, pickup/shipping label emails, reusable templates, and Excel/CSV exports.
- Designed secured REST APIs with AWS Cognito/JWT authentication, centralized exception handling, typed response wrappers, multi-schema persistence, and structured JSON logging.
- Developed and maintained domain-specific modules covering Client Management, Customer Profiles, Billing, Claims, and Transit Times, supporting customer configuration and transportation operations.
- Integrated additional external services, including Google Geocoding, NMFTA, and Shiplify, for geocoding and transportation-related workflows.
- Implemented document storage, billing document workflows, claims document uploads, PDF generation, and email notifications using AWS S3 and SES.
- Deployed serverless AWS Lambda infrastructure using Mangum and AWS CDK, with GitHub Actions CI/CD, Pytest, Ruff, Mypy, and 80%+ test coverage.

Ideyalabs **Hyderabad, India**
Backend Developer Intern *2024–2025*

HLF – Blockchain Transaction System

- Implemented RESTful APIs in Go for blockchain transaction management including issue, redeem, and transfer operations.
- Engineered Go-based smart contracts (chaincode) for Hyperledger Fabric to enforce business logic across multiple workflows.
- Led migration from Corda-based blockchain to Hyperledger Fabric, improving performance and interoperability.

Projects

Data Agent:

- Developed an AI-powered data agent using Python, LangGraph, LangChain, and Pydantic to route natural language requests between ETL and SQL analysis workflows.
- Built ETL workflows to extract data from APIs and files, transform datasets using Pandas, and export results in CSV, JSON, and Parquet formats.
- Implemented database analysis capabilities with SQLAlchemy and SQL, enabling schema inspection and LLM-assisted SQL generation and safety checks.
- Designed modular agent workflows with specialized tools for data extraction, transformation, loading, and database querying through natural language.

Chatbot:

- Engineered a full-stack AI chat application with React + Tailwind CSS frontend and FastAPI backend supporting multi-threaded conversations with PostgreSQL-backed persistent history.
- Integrated OpenAI GPT-4 with LangGraph for stateful conversation management and custom tools, including Tavily web search for real-time information retrieval.
- Designed a service-layer architecture using SQLAlchemy ORM with proper session management for handling concurrent conversations.

Budget Buddy:

- Built an AI-powered command-line finance assistant using LangGraph and LangChain with intelligent financial tracking capabilities and a conversational AI interface.
- Created custom financial tools supporting income/expense tracking with date range analytics, real-time balance monitoring, and web search integration.
- Implemented a modular architecture with PostgreSQL, SQLAlchemy ORM, dependency injection, and persistent conversation memory.

Fitness AI Platform:

- Designed a microservices platform using Spring Boot with service discovery (Eureka), API gateway (Spring Cloud Gateway), and centralized configuration management.
- Implemented an event-driven architecture using RabbitMQ for asynchronous inter-service communication between user, activity, and AI recommendation services.
- Built an AI recommendation service using Google Gemini API with Spring WebFlux for reactive programming, MongoDB storage, and OAuth2 authentication.

ClearBG:

- Implemented a full-stack image background removal application using Spring Boot backend, React frontend, and ClipDrop API integration.
- Integrated Clerk authentication with webhook synchronization for user management and JWT-based API security using custom filters.
- Added a credit-based payment system using Razorpay integration and reactive Spring WebClient for asynchronous AI service communication.

Travel Agent AI:

- Developed an AI travel assistant using LangGraph with FastAPI backend, providing intelligent flight and hotel search capabilities through natural language queries.
- Integrated Pinecone vector database for conversation context retrieval and implemented custom tools for SerpAPI-based search with structured JSON responses.
- Designed an agent-based architecture with conversation summarization, PostgreSQL for data persistence, and LangChain for tool orchestration.

For more projects, visit: github.com/Akhil351

Achievements

LeetCode: Solved 530+ problems across Easy, Medium, and Hard difficulty levels covering arrays, trees, graphs, recursion, and dynamic programming.

Red Hat Certified: Enterprise Application Developer (EX183) – Validated expertise in Java EE application development, RESTful services, CDI, JPA, and JBoss EAP deployment.

Hackathon Finalist: Google Cloud Agentic AI Day (Hack2skill, 2025) – Ranked in the top 700 out of 57,000+ participants. Built an AI-powered monitoring system using Vertex AI, Gemini, Next.js, and ESP32 hardware within a 30-hour timeframe.